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Inequalities for Plane Partitions

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Abstract. Inequalities are important features in the context of sequences of numbers and polynomials. The Bessenrodt–Ono inequality for partition numbers and Nekrasov–Okounkov polynomials has only recently been discovered. In this paper we study the log-concavity (Turán inequality) and Bessenrodt–Ono inequality for plane partitions and their polynomization.

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Keywords. Inequalities, Plane Partitions, Polynomials.

1. Introduction and Main Results

In this paper we address inequalities for plane partitions and their polynomization. Plane partitions are, according to Stanley, fascinating generalizations of partitions of integers [33, Section 7.20]. Andrews [2] gave an excellent introduction of plane partitions in the context of higher-dimensional partitions. We also refer to Krattenthaler's survey on plane partitions in the work of Stanley and his school [21].

A plane partition π of n is an array $\pi = (\pi_{ij})_{i,j\geq 1}$ of non-negative integers π_{ij} with finite sum $|\pi| := \sum_{i,j\geq 1} \pi_{ij} = n$, which is weakly decreasing in rows and columns. It can be considered as the filling of a Ferrers diagram with weakly decreasing rows and columns, where the sum of all these numbers is equal to n. Let the numbers in the filling represent the heights for stacks of blocks placed on each cell of the diagram (Fig. 1). This is a natural generalization of the concept of classical partitions [2,30].

Let $p(n)$ denote the number of partitions of n and $\mathrm{pp}(n)$ the number of plane partitions of n. As usual, $p(0) := 1$ and $\mathrm{pp}(0) := 1$. In Table 1 we have listed the first values of the partition and plane partition function.

We investigate the log-concavity [6,32] and the Bessenrodt–Ono inequality [5,19] of plane partitions. A sequence of real, non-negative numbers

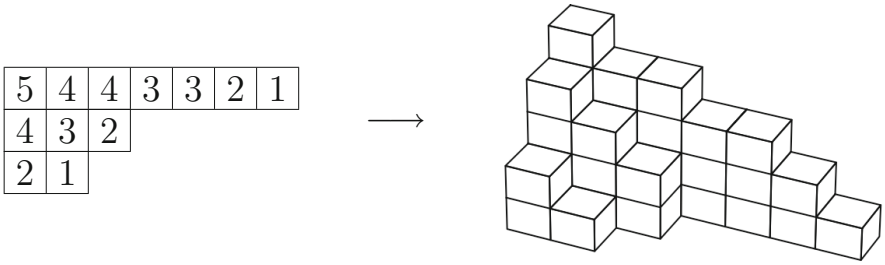


FIGURE 1. Representation of plane partitions

TABLE 1. Values for $0 \leq n \leq 10$

n	0	1	2	3	4	5	6	7	8	9	10
$p(n)$	1	1	2	3	5	7	11	15	22	30	42
$pp(n)$	1	1	3	6	13	24	48	86	160	282	500

$\{\alpha(n)\}_{n=0}^{\infty}$ is log-concave if for all $n \in \mathbb{N}$:

$$\alpha(n)^2 > \alpha(n-1)\,\alpha(n+1). \tag{1.1}$$

We say log-concave at n if (1.1) is satisfied for a specific n. A sequence of real non-negative numbers $\{\alpha(n)\}_{n=0}^{\infty}$ satisfies the Bessenrodt–Ono inequality for a set $S \subset \mathbb{N} \times \mathbb{N}$, if for all $(a, b) \in S$:

$$\alpha(a)\,\alpha(b) > \alpha(a+b). \tag{1.2}$$

1.1. Related Work and Recent Results

We recall that Nicolas [29] proved that the partition function is log-concave for $n > 25$:

$$p(n)^2 > p(n-1)\,p(n+1). \tag{1.3}$$

It is valid for all n even and fails for $1 \leq n \leq 25$ odd. In the course of proving several conjectures of Chen and Sun, DeSalvo and Pak [11] reproved this result. They remarked that due to the Hardy–Ramanujan asymptotic formula

$$p(n) \sim \frac{1}{4\sqrt{3n}}\, e^{\pi\sqrt{\frac{2}{3}n}} \quad \text{as } n \to \infty,$$

there is no way of knowing precisely when the asymptotic formula dominates the calculation. Their proof is based on Rademacher type estimates [31] by Lehmer (e. g. [22,23]), which provide the demanded, explicit, guaranteed error estimate. This proves the log-concavity for $n \geq 2,600$ [11]. More generally, let $p_k(n)$ be the k-colored partition function, obtained for every $k \in \mathbb{N}$ by the generating function

$$\prod_{n=1}^{\infty} (1-q^n)^{-k} = \sum_{n=0}^{\infty} p_k(n)\, q^n,$$

which is essentially the kth power of the reciprocal of the Dedekind eta function [30]. Then (1.3) was extended by Chern–Fu–Tang [9] to an interesting conjecture: Suppose $k, \ell, n \in \mathbb{N}$ with $k \geq 2$ and $n > \ell$. Let $(k, n, \ell) \neq (2, 6, 4)$. Then

$$p_k(n-1)\, p_k(\ell+1) \geq p_k(n)\, p_k(\ell)\,.$$

The conjecture was extended to $k \in \mathbb{R}_{\geq 2}$ [17], which involves the so-called D'Arcais polynomials or Nekrasov–Okounkov polynomials [13,28]. The Conjecture by Chern–Fu–Tang and some portion of the Conjecture by Heim–Neuhauser was recently proven by Bringmann, Kane, Rolen, and Tripp [7]. The proof is based on Rademacher type formulas for the coefficients of powers of the Dedekind eta function, utilizing the weak modularity property.

Bessenrodt and Ono [5] discovered a beautiful and simple inequality for partition numbers. Let $a, b \in \mathbb{N}$. Suppose $a, b \geq 2$ and $a + b \geq 10$, then

$$p(a)\, p(b) > p(a+b). \tag{1.4}$$

The inequality is symmetric and always fails for $a = 1$ or $b = 1$. Let $2 \leq b \leq a$. There is equality for the pairs $(4,3), (6,2), (7,2)$ and the opposite inequality of (1.4) is exactly true for the pairs $(2,2), (3,2), (4,2), (5,2), (3,3), (5,3)$. Bessenrodt and Ono's proof is based on an analytic result of Lehmer [23] of Rademacher type, similar to the proof of the log-concavity of $p(n)$ [11,29]. Shortly after the result was published, Alanazi, Gangola III, and Munagi [1] came up with a subtle combinatorial proof. Chern, Fu, and Tang [9] generalized and proved the Bessenrodt–Ono inequality to k-colored partitions. In [19] this was extended to k real, again involving polynomials. The proof was given by induction and involving derivatives. Further, the work of Bessenrodt and Ono triggered the results of Beckwith and Bessenrodt [3] on k-regular partitions, Hou and Jagadeesan [20] on the numbers of partitions with ranks in a given residue class modulo 3 and Males [27] for general t, and Heim and Neuhauser [16], and Dawsey and Masri [10] for the Andrews *spt*-function.

We performed several numerical experiments and are convinced that some of the recorded results can be transferred to plane partitions $\mathrm{pp}(n)$ and its generalization. MacMahon [24–26] proved the following non-trivial result, which took him several years. The generating function of the plane partition is given by

$$\prod_{n=1}^{\infty} (1-q^n)^{-n} = \sum_{n=0}^{\infty} \mathrm{pp}(n)\, q^n. \tag{1.5}$$

Since this generating function is not related to a weakly modular form, in contrast to the partition numbers, we have only an asymptotic formula provided by Wright [34] based on the circle and saddle point method for plane partition numbers. Wright proved the following asymptotic behavior as n approaches infinity:

$$\mathrm{pp}(n) \sim \frac{\zeta(3)^{\frac{7}{36}}}{\sqrt{12\pi}} \left(\frac{2}{n}\right)^{\frac{25}{36}} \exp\left(3\,\zeta(3)^{\frac{1}{3}} \left(\frac{n}{2}\right)^{\frac{2}{3}} + \zeta'(-1)\right). \tag{1.6}$$

Here $\exp(z) = \mathrm{e}^z$ and $\zeta(s)$ denotes the Riemann zeta function. Thus, we are in a similar situation as described before by DeSalvo and Pak [11] for partition numbers.

Recently, we invented a new proof method [18] and reproved some known results related to the Bessenrodt–Ono inequality for the partition function, the k-colored partitions and extension to the D'Arcais polynomials.

1.2. Main Results: Plane Partitions

We first start with the Bessenrodt–Ono inequality (1.2).

Theorem 1.1. (Bessenrodt–Ono inequality) *Let a and b be positive integers. Let $a, b \geq 2$ and $a + b \geq 12$. Then*

$$\mathrm{pp}(a)\,\mathrm{pp}(b) > \mathrm{pp}(a+b).$$

Equality is never satisfied.

Due to symmetry, let us assume that $2 \leq b \leq a$. Then $\mathrm{pp}(a)\,\mathrm{pp}(b) < \mathrm{pp}(a+b)$ for $(a,b) \in \{(a,2) : 2 \leq a \leq 9\} \cup \{(a,3) : 2 \leq a \leq 5\}$. Note that $\mathrm{pp}(n) < \mathrm{pp}(n+1)$, similar to $p(n) < p(n+1)$. We remark for $1 \leq n \leq 3$ the property

$$(\mathrm{pp}\,(n))^2 < \mathrm{pp}\,(2n)$$

and for all other n the opposite. Based on our investigations we state the following:

Conjecture 1. *Let $n \geq 12$. Then the sequence $\{\mathrm{pp}(n)\}_n$ of plane partitions is log-concave:*

$$\mathrm{pp}(n)^2 > \mathrm{pp}(n-1)\,\mathrm{pp}(n+1). \tag{1.7}$$

We can show with Wright's formula (1.6), that (1.7) is true for large n, and it seems that this is already true for all n even and for all $n \geq 12$.

Theorem 1.2. (Log-Concavity) *Let $12 \leq n \leq 10^5$. Then* Conjecture 1 *is true. It is further true for all n even and false for all odd n below 12. Furthermore there is an N such that it is true for all $n > N$.*

It would be very interesting to determine such an N of reasonable size and to finally prove the conjecture.

1.3. Main Results: Polynomization

It is possible to consider $\{\mathrm{pp}(n)\}_n$ as special values of a family of polynomials $\{P_n(x)\}_n$. We will have $\mathrm{pp}(n) = P_n(1)$. This makes it possible to generalize the Bessenrodt–Ono inequality and the log-concavity. We view the inequalities as a property of the largest positive real zeros of new polynomials associated with $\{P_n(x)\}_n$.

Definition. Let $\sigma_2(n) := \sum_{d|n} d^2$. Let $P_0(x) := 1$ and

$$P_n(x) := \frac{x}{n} \sum_{k=1}^{n} \sigma_2(k)\, P_{n-k}(x). \tag{1.8}$$

TABLE 2. Polynomials $P_n(x)$ for $n \in \{1,2,3,4,5\}$

n	$P_n(x)$
1	x
2	$\frac{1}{2}x^2 + \frac{5}{2}x$
3	$\frac{1}{6}x^3 + \frac{5}{2}x^2 + \frac{10}{3}x$
4	$\frac{1}{24}x^4 + \frac{5}{4}x^3 + \frac{155}{24}x^2 + \frac{21}{4}x$
5	$\frac{1}{120}x^5 + \frac{5}{12}x^4 + \frac{115}{24}x^3 + \frac{163}{12}x^2 + \frac{26}{5}x$

We have listed the first polynomials in Table 2.

Let $q, z \in \mathbb{C}$ and $|q| < 1$. It is a standard procedure to show that

$$\sum_{n=0}^{\infty} P_n(z)\, q^n = \prod_{n=1}^{\infty} (1-q^n)^{-n\,z} = \exp\left(z \sum_{n=1}^{\infty} \sigma_2(n)\, \frac{q^n}{n} \right). \tag{1.9}$$

Note that the second equality follows as

$$\begin{aligned} \ln\left(\prod_{n=1}^{\infty} (1-q^n)^{-nz} \right) &= \sum_{n=1}^{\infty} -nz \ln(1-q^n) = z \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \frac{n}{k} q^{nk} \\ &= z \sum_{n=1}^{\infty} \sum_{k|n} \frac{k^2}{n} q^n = z \sum_{n=1}^{\infty} \frac{\sigma_2(n)}{n} q^n. \end{aligned}$$

Deriving the last expression in (1.9) shows that the coefficients of its power series exactly fulfill (1.8). Thus,

$$n\, \mathrm{pp}(n) = \sum_{k=1}^{n} \sigma_2(k)\, \mathrm{pp}(n-k), \tag{1.10}$$

applying MacMahon's discovery (1.5). Let k be a positive integer. We would like to call $P_n(k)$ the *k-colored* plane partitions (compare [4]), but at the moment there is no combinatorial interpretation available, as in the case of partitions [7,19]. The topic is quite complicated, since MacMahon's result is already non-trivial and $\mathrm{pp}(n)$ can also be identified with the number of all partitions of $n = n_1 + n_2 + \cdots$, where each part n_j is allowed to have n_j colors.

1.3.1. Bessenrodt–Ono Inequalities.

Theorem 1.3. *Let $x \in \mathbb{R}$ and $x > 5$. Then*

$$P_a(x)\, P_b(x) > P_{a+b}(x)$$

for all positive integers a and b.

It is also possible to get results for $x = 1,2,3,4,5$. This leads to restrictions on a and b, reflected in Table 3, where we have recorded the largest real zero of

$$P_{a,b}(x) := P_a(x)\, P_b(x) - P_{a+b}(x).$$

TABLE 3. Approximative largest real zeros of $P_{a,b}(x)$ for $1 \leq a, b \leq 12$

a/b	1	2	3	4	5	6	7	8	9	10	11	12
1	5.0	3.2	3.0	2.6	2.5	2.3	2.2	2.1	2.1	2.0	2.0	1.9
2	3.2	1.9	1.6	1.4	1.3	1.2	1.1	1.1	1.0	1.0	1.0	0.9
3	3.0	1.6	1.5	1.2	1.1	1.0	1.0	0.9	0.9	0.8	0.8	0.8
4	2.6	1.4	1.2	0.9	0.9	0.8	0.7	0.7	0.6	0.6	0.6	0.6
5	2.5	1.3	1.1	0.9	0.9	0.7	0.7	0.6	0.6	0.6	0.6	0.5
6	2.3	1.2	1.0	0.8	0.7	0.6	0.6	0.5	0.5	0.5	0.5	0.4
7	2.2	1.1	1.0	0.7	0.7	0.6	0.6	0.5	0.5	0.5	0.5	0.4
8	2.1	1.1	0.9	0.7	0.6	0.5	0.5	0.4	0.4	0.4	0.4	0.4
9	2.1	1.0	0.9	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.4	0.3
10	2.0	1.0	0.8	0.6	0.6	0.5	0.5	0.4	0.4	0.4	0.3	0.3
11	2.0	1.0	0.8	0.6	0.6	0.5	0.5	0.4	0.4	0.3	0.3	0.3
12	1.9	0.9	0.8	0.6	0.5	0.4	0.4	0.4	0.3	0.3	0.3	0.3

Thus, studying the polynomials $P_n(x)$ and $P_{a,b}(x)$ and their leading coefficients and zeros, provides the big picture and reveals information on the original task, studying properties of plane partitions $\mathrm{pp}(n) = P_n(1)$. Note that $P_{a,b}(x)$ goes to infinity for $a, b \geq 1$, as x goes to infinity.

Theorem 1.4. *Let $x \in \mathbb{R}$ and $x \geq 2$. Then*

$$P_a(x)\, P_b(x) > P_{a+b}(x)$$

for all positive integers a and b satisfying $a + b \geq 12$.

It would be interesting to search for a combinatorial proof for the plane partition numbers (Theorem 1.1) and their generalization to k-colored plane partitions (Theorems 1.3 and 1.4).

1.3.2. Log-Concavity. In the spirit of the Chern–Fu–Tang Conjecture [9] on k-colored partitions and their polynomization [17] (see also [7]), we consider the polynomials

$$\Delta_{a,b}(x) := P_{a-1}(x)P_{b+1}(x) - P_a(x)\, P_b(x).$$

Note that $\Delta_{a+1,a-1}(x) = P_a(x)^2 - P_{a-1}(x)\, P_{a+1}(x)$. Then $\Delta_{a+1,a-1}(1) > 0$ is the log-concavity condition for $\mathrm{pp}(a)$.

This leads to the following conjecture on the log-concavity of the polynomials $P_n(x)$. For orthogonal polynomials, such kinds of inequalities are called Turán inequalities.

Conjecture 2. (Turán inequality) *Let a be an integer with $a \geq 12$ and let x be a real number with $x \geq 1$. Then*

$$P_a(x)^2 > P_{a-1}(x)\, P_{a+1}(x).$$

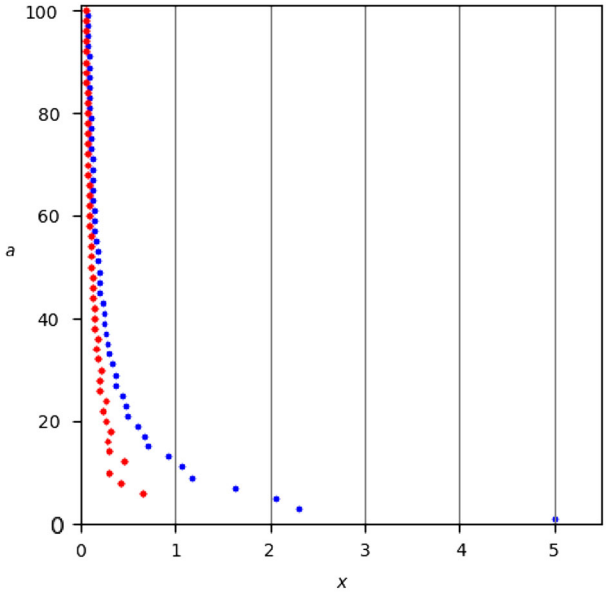


FIGURE 2. Zeros of $\Delta_{a+1,a-1}(x)$ with positive real part for $1 \leq a \leq 100$. Blue labels the real zeros and red the non-real zeros (color figure online)

This is a natural extension of Conjecture 1 on the log-concavity of plane partitions.

It would also be interesting to study the hyperbolicity of the associated Jensen polynomials [12] and higher Turán inequalities [8]. Let $\alpha(m) := \mathrm{pp}(m)$ or more generally, $\alpha(m) := P_m(x)$, $x \in \mathbb{R}_{>0}$. The Jensen polynomial of degree d and shift n attached to the sequence $\{\alpha(0), \alpha(1), \alpha(2), \ldots\}$ of non-negative real numbers is the polynomial

$$J_{\alpha}^{d,n}(X) := \sum_{k=0}^{d} \binom{d}{k} \alpha(n+k)\, X^k.$$

It follows from the zero distribution of the polynomial $\Delta_{a+1,a-1}(x) := P_a(x)^2 - P_{a-1}(x)\, P_{a+1}(x)$ (see Fig. 2) that Conjecture 2 is true for $12 \leq a \leq 100$, since the polynomial goes to infinity, as x goes to infinity. Let a be even and $2 \leq a \leq 1000$, then the Turán inequality in Conjecture 2 is already valid for $x > 0$. Actually, in this case all coefficients of $\Delta_{a+1,a-1}(x)$ are non-negative.

The generalization of the former Chern–Fu–Tang conjecture of k-colored partitions and its polynomization [17] leads to:

Conjecture 3. *Let a and b be integers. Suppose $a - 1 > b \geq 0$ and $(a, b) \notin \{(4, 2), (6, 4)\}$. Then for all real numbers $x \geq 2$:*

$$\Delta_{a,b}(x) := P_{a-1}(x)\, P_{b+1}(x) - P_a(x)\, P_b(x) > 0.$$

TABLE 4. Values of $\sigma_2(n)$ for $n \in \{1,2,3,4,5\}$

n	1	2	3	4	5
$\sigma_2(n)$	1	5	10	21	26

We refer to Table 8, where we recorded the largest real zero for $\Delta_{a,b}$ for all pairs (a,b) with $a-1 > b \geq 0$ and $2 \leq a \leq 19$, which shows that the Conjecture 3 is valid for all admissible pairs (a,b) in this range.

2. Basic Properties of $\sigma_2(n)$ and $P_n(x)$

For proof of Theorems 1.1 and 1.3 we need some elementary properties of $\sigma_2(n)$ and $P_n(x)$. As a further evidence that Conjecture 2 most likely holds true for even arguments n, we prove that $\sigma_2(n)/n$ is log-concave for n even. Note that this is false for $\frac{1}{n}\sum_{d|n} d$, where n is even (which is related to the Chern–Fu–Tang conjecture for k-colored partitions).

Proposition 2.1. *Let n be an even, positive integer. Then*

$$\left(\frac{\sigma_2(n)}{n}\right)^2 > \frac{\sigma_2(n-1)}{n-1}\,\frac{\sigma_2(n+1)}{n+1}.$$

Proof. For n odd we estimate

$$\begin{aligned}\sigma_2(n) &= \sum_{t|n} t^2 = \sum_{t|n}\left(\frac{n}{t}\right)^2 = n^2\sum_{t|n} t^{-2}\\ &\leq n^2\left(1-\frac{1}{4}\right)\sum_{t} t^{-2} = \frac{3n^2}{4}\zeta(2) = \frac{\pi^2}{8}n^2 < \frac{5}{4}n^2.\end{aligned}$$

Note that the first inequality follows by just taking all odd integers t instead of only divisors of n and the second one as $\pi^2 < 10$.

Now, let $n \geq 6$ be even. Then $\sigma_2(n) \geq n^2 + \left(\frac{n}{2}\right)^2 + 4 + 1 = \frac{5}{4}\left(n^2+4\right)$. Therefore,

$$\frac{\sigma_2(n-1)\,\sigma_2(n+1)}{n^2-1} < \frac{25}{16}\left(n^2-1\right) < \left(\frac{5}{4}\frac{n^2+4}{n}\right)^2 \leq \left(\frac{\sigma_2(n)}{n}\right)^2.$$

The inequality holds also for even $n \leq 4$ (Table 4). □

Note, that $\sigma_2(n) < \tilde{\sigma}_2(n) := 2\,n^2$ for all $n \in \mathbb{N}$. This follows from

$$\sum_{t|n} t^2 = n^2\sum_{t|n}\frac{1}{t^2} \leq n^2\left(1+\int_1^n t^{-2}\,\mathrm{d}t\right) < 2n^2.$$

The functions $P_n(x)$ are polynomials of degree n. This can be deduced directly from the recurrence formula (1.8). We have $P_n(x) = \frac{1}{n!}\sum_{m=1}^{n} A_{n,m}\,x^m$, where $A_{n,m} \in \mathbb{N}$ for $n \geq 1$. We later use the fact that $A_{n,1} = n!\,\frac{\sigma_2(n)}{n}$. Further, we have the following properties.

Proposition 2.2. *Let n, m be natural numbers and $x \geq 1$ real. Then*

$$P_n'(x) = \sum_{k=1}^{n} \frac{\sigma_2(k)}{k} P_{n-k}(x), \tag{2.1}$$

$$P_{n+1}(x) > P_n(x), \tag{2.2}$$

$$P_n(x) > \sum_{\ell=1}^{m} \frac{1}{\ell!} \binom{n+\ell-1}{2\ell-1} x^\ell \text{ for } n > 1. \tag{2.3}$$

Proof. The formula for the derivative $P_n'(x)$ is given similarly as that obtained for the polynomials associated with $\sigma(n) = \sum_{d|n} d$ [15].

We prove (2.2) by induction. Let $\Delta_n(x) := P_n(x) - P_{n-1}(x)$. Let $n = 1$. Then we have $\Delta_1(x) = x - 1 > 0$ for all $x > 1$. Suppose $\Delta_m(x) > 0$ for all $1 \leq m \leq n$. We obtain for $P_{n+1}'(x)$ the strict lower bound

$$\sum_{k=1}^{n} \frac{\sigma_2(k)}{k} P_{n+1-k}(x) \geq \sum_{k=1}^{n} \frac{\sigma_2(k)}{k} P_{n-k}(x).$$

Thus, $P_{n+1}'(x) > P_n'(x)$. The plane partition function is strictly increasing:

$$\mathrm{pp}(n) < \mathrm{pp}(n+1)$$

for all $n \in \mathbb{N}$, since every plane partition of n can be lifted to a plane partition of $n+1$. This provides

$$P_{n+1}(1) = \mathrm{pp}(n+1) > \mathrm{pp}(n) = P_n(1).$$

Now $P_{n+1}(x) > P_n(x)$ for $x = 1$ and since we have shown that $P_{n+1}'(x) > P_n'(x)$ it also holds for $x > 1$. Thus, the claim is proven.

The lower bound given in (2.3) is obtained in the following way. We have $n^2 < \sigma_2(n)$ for $n > 1$ and $\sigma_2(1) = 1$. Thus, the coefficients of the polynomial $S_n(x)$, defined by $S_0(x) := 1$ and $S_n(x) = \frac{x}{n} \sum_{k=1}^{n} k^2 S_{n-k}(x)$, are smaller than the coefficients of $P_n(x)$ for $n > 1$. The mth coefficient is given by $\frac{1}{m!}\binom{n+m-1}{2m-1}$. This can be deduced from [14, Section 4]. Comparing the size of the coefficients and the assumption that $x \geq 1$ proves the claim (2.3). □

Corollary 2.3. *Let $n \in \mathbb{N}$ and $x \geq 1$. Then $\Delta_n'(x) > 0$.*

3. Proof Strategy for the Bessenrodt–Ono Inequality for Plane Partitions

In this section we lay out a general strategy for proving Bessenrodt–Ono type inequalities. This also makes the appearance of exceptions transparent.

3.1. Input Data and Proof by Induction

Let $A, B, N_0 \in \mathbb{N}$ be given with $2 \leq B < N_0$ and $x_0 \in \mathbb{R}_{\geq 1}$. Let $n \geq B$. Let $\mathrm{S}(n)$ be the following mathematical statement for one of the three cases: $x \in \mathbb{R}$ with $x = x_0$, $x > x_0$, or $x \geq x_0$. For all $A \leq b \leq a$ with $a + b = n$ and all x with fixed case, we have

$$P_{a,b}(x) := P_a(x) P_b(x) - P_{a+b}(x) > 0.$$

Note that $P_{a,b}(x_0) = P_{b,a}(x_0)$ and that $\lim_{x\to\infty} P_{a,b}(x) = +\infty$.

Given input data A, B, x_0 by induction we prove $\mathrm{S}(n)$ for one of the given cases. We choose N_0 and prove first manually or by numerical calculation (utilizing PARI/GP) that $\mathrm{S}(\ell)$ is true for all $B \le \ell \le N_0$. In the case of $x \ge x_0$ or $x > x_0$ we study the real zeros of the polynomial $P_{a,b}(x)$ for all $A \le b \le a$ with $a + b = \ell$. Let $n \ge N_0$. Then we prove $\mathrm{S}(n)$ by assuming that $\mathrm{S}(m)$ is true for all $B \le m \le n-1$, the induction hypothesis.

3.2. Basic Decomposition

Let $A, B \in \mathbb{N}$ be given with $B \ge 2$. Further, let $a, b \in \mathbb{N}$ satisfy $A \le b \le a$ and $a + b \ge B$. We define

$$k_0 := a - \max\{B - b, A\} + 1,$$
$$f_k(a,b,x) := \sigma_2(k)\left(\frac{P_{a-k}(x)\,P_b(x)}{a} - \frac{P_{a+b-k}(x)}{a+b}\right).$$

We consider the decomposition $P_{a,b}(x) = L_{a,b}(x) + R_{a,b}(x)$, where

$$L = L_{a,b}(x) := -\sum_{k=1}^{b} \frac{\sigma_2(k+a)}{a+b} P_{b-k}(x),$$
$$R = R_{a,b}(x) := \sum_{k=1}^{a} f_k(a,b,x).$$

Utilizing (2.2) leads immediately to

$$L > -4\,b\,a\,P_b(x).$$

Further, let $R = R_1 + R_2 + R_3$, where

$$R_1 := f_1(a,b,x),\ R_2 := \sum_{k=2}^{k_0-1} f_k(a,b,x),\ R_3 := \sum_{k=k_0}^{a} f_k(a,b,x).$$

Suppose $P_{a+b-k}(x) < P_{a-k}(x)\,P_b(x)$ for $1 \le k \le k_0 - 1$, then

$$R_1 > \frac{b}{2\,a^2} P_{a-1}(x)\,P_b(x) \text{ and } R_2 > 0.$$

Further, we put $R_3 = R_{31} + R_{32} + R_{33}$, where

$$R_{31} := \sum_{k=k_0}^{a-A} f_k(a,b,x),\ R_{32} := \sum_{k=a-A+1}^{a-1} f_k(a,b,x),\ R_{33} := f_a(a,b,x).$$

Then $R_{33} > 0$ and $R_{32} = 0$ if $A = 1$. Moreover, let $B - b \le A$, then $R_{31} = 0$. Thus, R_3 can only have a negative contribution to R if $A \ge 2$ (by R_{32}) and if $B - b > A$ (by R_{31}). Note that $n^2 \le \sigma_2(n) < \tilde{\sigma}_2(n) := 2\,n^2$. We obtain by straight-forward estimations the following.

Lemma 3.1. *Let $B - b > A$. Then*

$$R_{31} > \frac{a - A - k_0 + 1}{a}\left(k_0^2\,P_A(x)\,P_b(x) - \tilde{\sigma}_2(a-A)P_{a-b-k_0}(x)\right).$$

If we know that some $P_k(x)\,P_b(x) - P_{b+k}(x)$ *is negative we can use the estimate*

$$R_{31} > \frac{\tilde{\sigma}_2(a)}{a} \sum_{k=A}^{B-b} \min\left\{0, P_k(x)\,P_b(x) - P_{k+b}(x)\right\}. \tag{3.1}$$

4. Proof of Theorems 1.3 and 1.4

Proof of Theorem 1.3. Let $x > 5$. We prove that $P_{a,b}(x) > 0$ for all $a, b \geq 1$. Due to symmetry, we can assume $b \leq a$. We follow the strategy presented in Sect. 3. Let $A = 1$ and $B = 2$. Let $x_0 = 5$ and $x > x_0$. These refer to Sect. 3.1. Note that $x = x_0$ does not work, since $P_{1,1}(5) = 0$. Let $N_0 = 12$. Then the mathematical statement $\mathrm{S}(n)$ is true for all $A \leq a, b \leq N_0$ (see Table 3). Let $n > N_0$, we assume that $\mathrm{S}(m)$ is true for all $B \leq m \leq n-1$. Let $a + b = n$ with $A \leq b \leq a$. Then $P_{a,b}(x) > L + R_1$. Note that $R_2, R_3 \geq 0$. With (2.2) we have

$$L > -4\,a\,b\,P_b(x),$$
$$R > \frac{b}{2\,a^2} P_{a-1}(x)\,P_b(x).$$

Final step. Putting things together and estimating $P_{a-1}(x)$ from below by (2.3), we obtain

$$P_{a,b}(x) > \frac{b\,P_b(x)}{2\,a^2}\left(-8\,a^3 + \sum_{\ell=1}^{5} \binom{a+\ell-2}{2\,\ell-1} \frac{5^\ell}{\ell!}\right). \tag{4.1}$$

The expression in brackets on the right hand side of (4.1) is polynomial in a of degree 9 with a positive leading coefficient. Calculating the largest real zero ≈ 6.95 with PARI/GP shows that the right hand side is positive for $a \geq 7$. □

Proof of Theorem 1.4. Let $x \geq 2$. We put $A = 1$ and $B = 12$. Let $N_0 = 58$. These refer again to Sect. 3.1. Then by induction, as before, we obtain

$$P_{a,b}(x) > L + R_1 + R_{31},$$

where for $b \geq 12$ we can apply the induction hypothesis and obtain $R_{31} > 0$. In case $1 \leq b \leq 11$ we find from Table 3 that $P_{a-k,b}(2) > 0$ for $b = 1$ and $a - k \notin \{1, 2, \ldots, 11\}$ or $k = a - 1$ and $b \notin \{1, 2, \ldots, 11\}$. Therefore,

$$R_{31} > \sum_{k=a-11}^{a-1} \frac{\sigma(k)\,P_{a-k,1}(x)}{a}.$$

It can be checked that the polynomials $P_{k,1}(x)$, $1 \leq k \leq 10$, are monotonically increasing for $x \geq 2$. Indeed, the leading coefficient of $P_{k,1}(x)$ is $\frac{1}{(k-1)!}$ and we have computed the largest real zero of $P'_{k,1}(x)$ with PARI/GP which are all less than 2 for $2 \leq k \leq 10$ and $P_{1,1}(x) > -4 = P_{1,1}(2) - 1$. Thus,

$$R > R_{31} > -642 \cdot 2a \geq -642\,a\,b\,P_b(x)$$

from (3.1) and Table 5.

TABLE 5. Values of $\sum_{k=1}^{11-b} \min\{0, P_k(2) P_b(2) - P_{k+b}(2)\}$ for $b \le 10$

b	1	2	3	4	5	6	7	8	9	10
	−641	−4	−11	−16	−38	−52	−101	−126	−180	−110

Final step: Putting everything together leads to

$$P_{a,b}(x) > \frac{b P_b(x)}{2a^2} \left(-1292a^3 + \sum_{\ell=1}^{8} \binom{a+\ell-2}{2\ell-1} \frac{2^\ell}{\ell!} \right). \tag{4.2}$$

In the last step we used the property (2.3) and that $x \ge 2$. We obtain that the expression (4.2) is positive for all $a \ge 27$. Since the leading coefficient of $P_{a,b}(x)$ is positive we only have to determine the largest real zero of all remaining $P_{a,b}(x)$. We checked this for $12 \le b + a \le 52$ with PARI/GP (compare Table 3). □

5. Proof of the Bessenrodt–Ono Inequality: Theorem 1.1

We start with the following auxiliary result.

Lemma 5.1. *We have* $\mathrm{pp}(2) = 3\,\mathrm{pp}(1)$. *Further, for* $n \ne 1$ *we have*

$$\mathrm{pp}(n+1) < 3\mathrm{pp}(n). \tag{5.1}$$

Proof. The proof is by mathematical induction. We checked with PARI/GP that (5.1) holds for $n \le N_0 = 24$. For the induction step we only use (1.10), (2.3), and the induction hypothesis.

Therefore, let $n > N_0$. Then

$$\begin{aligned} &3\mathrm{pp}(n) - \mathrm{pp}(n+1) \\ &= \frac{3}{n} \sum_{k=1}^{n} \sigma_2(k)\,\mathrm{pp}(n-k) - \frac{1}{n+1} \sum_{k=1}^{n+1} \sigma_2(k)\,\mathrm{pp}(n+1-k) \\ &= -\frac{\sigma_2(n+1)}{n+1} + \sum_{k=1}^{n} \sigma_2(k) \left(\frac{3}{n}\mathrm{pp}(n-k) - \frac{1}{n+1}\mathrm{pp}(n-k+1) \right). \end{aligned}$$

We now apply the induction hypothesis $\mathrm{pp}(n-k+1) < 3\mathrm{pp}(n-k)$ for each $1 \le k \le n$. Note that in these cases $0 \le n-k \le n-1 < n$. Since $\frac{1}{n} > \frac{1}{n+1}$ we drop all terms with $2 \le k \le n$ and only keep that with $k = 1$. Therefore, we obtain

$$\begin{aligned} &3\mathrm{pp}(n) - \mathrm{pp}(n+1) \\ &> -2\frac{(n+1)^2}{n+1} + \left(\frac{3}{n} - \frac{3}{n+1} \right) \mathrm{pp}(n-1) \end{aligned}$$

TABLE 6. Values of $\sum_{k=2}^{11-b} (\mathrm{pp}(k)\,\mathrm{pp}(b) - \mathrm{pp}(k+b))$ for $2 \le b \le 9$

b	2	3	4	5	6	7	8	9
	−106	−42	−17	−30	−16	−24	−20	−13

$$\ge -2(n+1) + \frac{3}{(n+1)\,n} \sum_{\ell=1}^{3} \binom{n+\ell-2}{2\ell-1} \frac{1}{\ell!} > 0$$

where we have used (2.3) with $x = 1$ and $m = 3$. The last inequality follows as the last expression if multiplied with $(n+1)\,n$ is a polynomial of degree 5 with leading coefficient $\frac{1}{240} > 0$ and the largest real zero computed with PARI/GP is < 25. □

Remarks. It would be good to have a combinatorial proof of Lemma 5.1.

Proof of Theorem 1.1. The proof is again by induction. We follow the proof strategy stated in Sect. 3. See also Sect. 4.

Let $A = 2$, $B = 12$ and $x = x_0 = 1$ referring to Sect. 3.1. Then $k_0 = a - \max\{B - b, A\} + 1 = a - \max\{11 - b, 1\}$. Furthermore, $L > -4ab\mathrm{pp}(b)$, $R_1 > \frac{b}{2a^2}\mathrm{pp}(a-1)\,\mathrm{pp}(b)$, and $R_2 > 0$. For R_{33} we obtain the lower bound 0 and for R_{32} we obtain

$$\sigma_2(a-1)\left(\frac{\mathrm{pp}(b)}{a} - \frac{\mathrm{pp}(b+1)}{a+b}\right) \ge -\sigma_2(a-1)\,\frac{2\mathrm{pp}(b)}{a} > -4a\mathrm{pp}(b).$$

Finally,

$$\begin{aligned} R_{31} &> \sum_{k=k_0}^{a-A} \frac{\sigma_2(k)}{a} \left(\mathrm{pp}(a-k)\,\mathrm{pp}(b) - \mathrm{pp}(a+b-k)\right) \\ &\ge -106 \cdot 2a = -212a \end{aligned}$$

(Table 6). Therefore, note $R_3 > -4a\mathrm{pp}(b) - 212a > -38ab\mathrm{pp}(b)$. Putting everything together leads to

$$\begin{aligned} \mathrm{pp}(a)\,\mathrm{pp}(b) - \mathrm{pp}(a+b) &> \frac{b\mathrm{pp}(b)}{2a^2}\left(-76a^3 + \mathrm{pp}(a-1)\right) \\ &> \frac{b\mathrm{pp}(b)}{2a^2}\left(-76a^3 + \sum_{\ell=1}^{3} \frac{1}{\ell!}\binom{a-\ell-2}{2\ell-1}\right) \end{aligned}$$

using (2.3). This is positive for $a \ge 237$. We have checked with PARI/GP that $\mathrm{pp}(a)\,\mathrm{pp}(b) - \mathrm{pp}(a+b) > 0$ for $B \le a+b \le N_0 = 472$. □

6. Proof of Theorem 1.2

Lemma 6.1. *Let $s \in \mathbb{R}$. There are $C_0, N > 0$ such that*

$$2n^s - (n+1)^s - (n-1)^s = (1-s)\,sn^{s-2} + E_n n^{s-3} \tag{6.1}$$

for $n > N$ with $|E_n| < C_0$.

Proof. We have

$$\begin{aligned}
&2n^s-(n-1)^s-(n+1)^s\\
&=\left(2-\left(1-\frac{1}{n}\right)^s-\left(1+\frac{1}{n}\right)^s\right)n^s\\
&=\left(2-\left(1-\frac{s}{n}-\frac{(1-s)\,s}{2n^2}+\frac{D_{1,n}}{n^3}\right)-\left(1+\frac{s}{n}-\frac{(1-s)\,s}{2n^2}+\frac{D_{2,n}}{n^3}\right)\right)n^s\\
&=(1-s)\,sn^{s-2}-E_nn^{s-3}
\end{aligned}$$

with $|D_{1,n}|,|D_{2,n}|,|E_n|<C_0$ for some $C_0,N>0$ and all $n>N$. □

Corollary 6.2. *Let $C_1>0$. There is $N>0$ such that*

$$\begin{aligned}
1+\frac{C_1}{9}n^{-4/3}&<\exp\left(2C_1n^{2/3}-C_1\left(n+1\right)^{2/3}-C_1\left(n-1\right)^{2/3}\right)\\
&<1+\frac{4C_1}{9}n^{-4/3}
\end{aligned}\tag{6.2}$$

for all $n>N$.

Proof. For example for the lower bound we obtain from (6.1)

$$2n^{2/3}-\left(n-1\right)^{2/3}-\left(n+1\right)^{2/3}\geq\frac{2}{9}n^{-4/3}-2C_0n^{-7/3}>\frac{1}{9}n^{-4/3}$$

for some $C_0>0$ and all $n>N$ for some N. Therefore,

$$\begin{aligned}
\exp\left(2C_1n^{2/3}-C_1\left(n-1\right)^{2/3}-C_1\left(n+1\right)^{2/3}\right)&>\exp\left(\frac{C_1}{9}n^{-4/3}\right)\\
&>1+\frac{C_1}{9}n^{-4/3}.
\end{aligned}$$

□

Theorem 6.3. *Let $C_1,C_2,C_3>0$, $r,\gamma_1,\gamma_2\in\mathbb{R}$, $\beta(n)=C_2n^r\mathrm{e}^{C_1n^{2/3}}$, and*

$$\left|\frac{\alpha(n)}{\beta(n)}-1-\gamma_1n^{-2/3}-\gamma_2n^{-4/3}\right|\leq C_3n^{-2}$$

for $n>N_0$ for some N_0. There is an $N\geq N_0$ such that $(\alpha(n))_{n\geq N_0}$ is log-concave for $n>N$.

Proof. Let $f_\pm(n)=\gamma_1n^{-2/3}+\gamma_2n^{-4/3}\pm C_3n^{-2}$. Then there is an $N_1\geq N_0$ such that

$$\left(1+f_-(n)\right)\beta(n)<\alpha(n)<\left(1+f_+(n)\right)\beta(n)$$

for all $n>N_1$. Therefore,

$$\begin{aligned}
&\left(\alpha(n)\right)^2-\alpha(n-1)\,\alpha(n+1)\\
&\quad>\left(\left(1+f_-(n)\right)\beta(n)\right)^2\\
&\qquad-\left(1+f_+(n-1)\right)\left(1+f_+(n+1)\right)\beta(n-1)\,\beta(n+1)
\end{aligned}$$

$$= (1+f_-(n))^2 \beta(n+1)\beta(n-1)$$
$$\cdot \left(\exp\left(2C_1 n^{2/3} - C_1(n-1)^{2/3} - C_1(n+1)^{2/3}\right)\right.$$
$$\left. - \frac{(1+f_+(n-1))(1+f_+(n+1))}{(1+f_-(n))^2}\left(\frac{n^2-1}{n^2}\right)^r\right).$$

Now

$$\frac{(1+f_+(n-1))(1+f_+(n+1))}{(1+f_-(n))^2}$$
$$= 1 + \frac{f_+(n-1)+f_+(n+1)-2f_-(n)}{1+f_-(n)}$$
$$+ \frac{(f_+(n-1)-f_-(n))(f_+(n+1)-f_-(n))}{(1+f_-(n))^2}.$$

With (6.1) we obtain

$$|f_+(n-1)+f_+(n+1)-2f_-(n)| = \left|-\frac{10}{9}\gamma_1 n^{-8/3} + E_n n^{-10/3}\right| < C_5 n^{-8/3}$$

with $|E_n| < C_4$ for some $C_4, C_5 > 0$, and

$$|(n+v)^u - n^u| = \left|1 + \frac{uv}{n} + \frac{U_n}{n^2} - 1\right| n^u = \left|uvn^{u-1} + U_n n^{u-2}\right| < C_6 n^{-5/3}$$

with $|U_n| < C_7$ for some $C_6, C_7 > 0$, and $u \le -2/3$. Therefore,

$$\left|\frac{(1+f_+(n-1))(1+f_+(n+1))}{(1+f_-(n))^2}\right| < 1 + C_8 n^{-8/3}$$

for some $C_8 > C_5$ which implies with (6.2)

$$\exp\left(2C_1 n^{2/3} - C_1(n-1)^{2/3} - C_1(n+1)^{2/3}\right)$$
$$- \frac{(1+f_+(n-1))(1+f_+(n+1))}{(1+f_-(n))^2}\left(1+\frac{1}{n^2-1}\right)^{-r}$$
$$> 1 + \frac{C_1}{9}n^{-4/3} - (1+C_8 n^{-8/3})\left(1+\frac{C_9}{n^2}\right) > 0$$

for some $C_9 > 0$. This is positive for $n > N$ and N sufficiently large. □

Proof of Theorem 1.2. Wright's formula ([34], Formula (2.21)) tells us that there are $C_1, C_2, C_3 > 0$, $r = -25/36$, and $\gamma_1, \gamma_2 \in \mathbb{R}$ such that

$$\mathrm{pp}(n) = C_2 n^{-\frac{25}{36}} \mathrm{e}^{C_1 n^{2/3}} \left(1 + \frac{\gamma_1}{n^{\frac{2}{3}}} + \frac{\gamma_2}{n^{\frac{4}{3}}} + \frac{G_n}{n^2}\right)$$

with $|G_n| < C_3$ for all $n > N_0$, N_0 sufficiently large. □

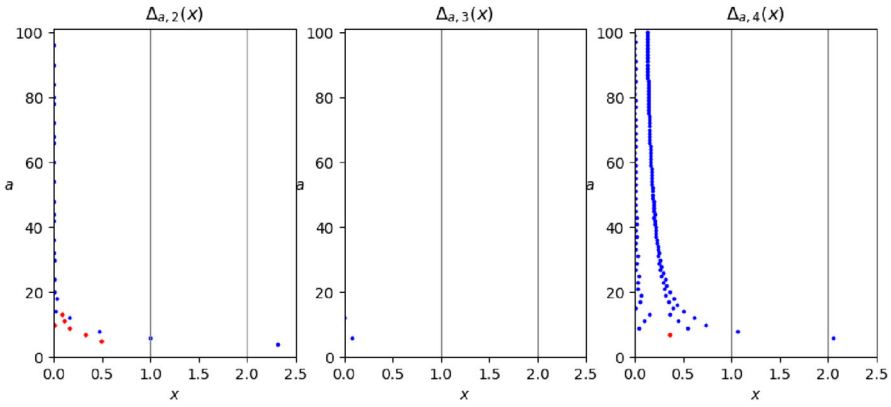


FIGURE 3. Zeros with the largest real part of $\Delta_{a,b}(x)$ for $2 \leq b \leq 4$, blue = real zero, red = non-real zero (color figure online)

7. Conjecture 2 and Conjecture 3

In this section we provide evidence for the Conjectures by information on the zeros of the underlying polynomials. Here we also use the crucial property that the leading coefficient of these polynomials always has positive sign.

7.1. Conjecture 2

Figure 2 indicates that it is most likely that if one considers the sequence of real parts of the zero of $\{\Delta_{a+1,a-1}(x)\}$ with the largest real part, that these numbers tend in the limit to zero. Another aspect is given by coefficients of these polynomials (Table 7). We observe that for a odd there are coefficients, which are negative. In the case a even, as already mentioned in the introduction, we have calculated the polynomials for $2 \leq a \leq 1000$ and observed that the coefficients are all non-negative. Let

$$\Delta_{a+1,a-1}(x) = \sum_{k=2}^{2a} B_{2a,k}\, x^k,$$

then we deduce from Proposition 2.1 that $B_{2a,2} > 0$.

7.2. Conjecture 3

Theorem 1.4 implies that $\Delta_{a,0}(x) = P_{a-1}(x)\, P_1(x) - P_a(x) > 0$ for $a \geq 12$ and $x \geq 2$. Since $\Delta_{a,0}(x) > 0$ also for $3 \leq a \leq 11$ (Table 8), we obtain:

Corollary 7.1. *Let $x \geq 2$. Then* Conjecture 3 *is valid for all $a \geq 3$ and $b = 0$.*

Next, we prove Conjecture 3 for $x \geq 1$ and all pairs $(a, 1)$ with $a \geq 3$. We refer also to Fig. 3 for the pairs (a, b) with $2 \leq b \leq 4$. Note, for $(a, 1)$ and $3 \leq a \leq 100$, there are no zeros with a positive real part.

Proposition 7.2. *Let $x \geq 1$ and $a \geq 3$. Then $\Delta_{a,1}(x) > 0$.*

TABLE 7. Polynomials $\Delta_{a+1,a-1}(x)$ for $a \in \{2,3,4,5\}$

a	$\Delta_{a+1,a-1}(x)$
2	$\frac{1}{12}x^4 + \frac{35}{12}x^2$
3	$\frac{1}{144}x^6 + \frac{5}{48}x^5 + \frac{145}{144}x^4 - \frac{101}{48}x^3 - \frac{145}{72}x^2$
4	$\frac{1}{2880}x^8 + \frac{1}{72}x^7 + \frac{67}{288}x^6 + \frac{137}{144}x^5 + \frac{11629}{2880}x^4 + \frac{1373}{144}x^3 + \frac{491}{48}x^2$
5	$\frac{1}{86400}x^{10} + \frac{1}{1152}x^9 + \frac{5}{192}x^8 + \frac{941}{2880}x^7 + \frac{63131}{28800}x^6 + \frac{6449}{1152}x^5 - \frac{1303}{864}x^4 - \frac{54613}{1440}x^3 - \frac{1671}{100}x^2$

TABLE 8. Approximative largest positive real zeros of $\Delta_{a,b}(x)$ for $0 \leq b < a-1 \leq 19$

$x_{a,b}$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
2	**5.00**																		
3	**3.16**	–																	
4	**3.00**	–	**2.31**																
5	**2.57**	–	–	–															
6	**2.50**	–	1.00	0.09	**2.05**														
7	**2.30**	–	–	–	–	–													
8	**2.25**	–	0.47	–	1.06	–	1.64												
9	**2.14**	–	–	–	0.55	–	0.73	–											
10	**2.09**	–	–	–	0.73	–	0.89	–	1.17										
11	**2.03**	–	–	–	0.45	–	0.55	–	–	–									
12	1.99	–	0.16	0.00	0.62	–	0.71	0.04	0.69	0.28	1.07								
13	1.94	–	–	–	0.36	–	0.46	–	–	–	–	–							
14	1.91	–	0.02	–	0.51	–	0.57	–	0.48	0.06	0.59	–	0.92						
15	1.88	–	–	–	0.39	–	0.46	–	0.27	–	0.33	–	0.46	–					
16	1.85	–	–	–	0.44	–	0.49	–	0.37	–	0.43	–	0.55	–	0.70				
17	1.82	–	–	–	0.35	–	0.41	–	0.19	–	0.25	–	0.33	–	–	–			
18	1.80	–	0.03	–	0.40	–	0.45	–	0.33	0.05	0.37	–	0.44	0.04	0.43	0.18	0.68		
19	1.78	–	–	–	0.32	–	0.38	–	0.17	–	0.22	–	0.29	–	–	–	–	–	
20	1.76	–	0.01	–	0.36	–	0.41	–	0.29	0.03	0.32	–	0.37	0.01	0.31	0.08	0.41	–	0.61

Proof. We deduce from Table 8 that $\Delta_{a,1}(x) > 0$ for $x > 0$ and $a \le 20$. Moreover, $\Delta_{a,1}(1) = 3\,\mathrm{pp}(a-1) - \mathrm{pp}(a) > 0$ for $a \ge 3$. This follows from Lemma 5.1. Let $\Delta_{a,1}(x) = x\,F_a(x)$. We prove that $F_a(x) > 0$ for $x \ge 1$ and $a \ge 3$ by induction on a. Let $a \ge 6$ and $F_m(x) > 0$ for $3 \le m < a$ and $x \ge 1$. We show that $F_a'(x) > 0$, which completes the proof, since $F_a(1) > 1$, for $a \ge 3$. Recall Formula (2.1). Then the derivative $F_a'(x)$ is equal to

$$\frac{x+5}{2}\sum_{k=1}^{a-1}\frac{\sigma_2(k)}{k}P_{a-1-k}(x) - \sum_{k=1}^{a}\frac{\sigma_2(k)}{k}P_{a-k}(x) + \frac{P_{a-1}(x)}{2}.$$

By the induction hypothesis we obtain

$$\frac{x+5}{2}\sum_{k=1}^{a-1}\frac{\sigma_2(k)}{k}P_{a-1-k}(x) > \frac{x+5}{2}\frac{\sigma_2(a-1)}{a-1} + \sum_{k=1}^{a-2}\frac{\sigma_2(k)}{k}P_{a-k}(x).$$

This leads to

$$\begin{aligned} F_a'(x) &> \frac{x+5}{2}\frac{\sigma_2(a-1)}{a-1} - \sum_{k=a-1}^{a}\frac{\sigma_2(k)}{k}P_{a-k}(x) + \frac{P_{a-1}(x)}{2} \\ &= \frac{P_{a-1}(x)}{2} - \frac{x\,\sigma_2(a-1)}{2(a-1)} + \frac{5\,\sigma_2(a-1)}{2\,(a-1)} - \frac{\sigma_2(a)}{a}. \end{aligned}$$

Moreover, we have

$$2\,F_a'(x) > \left(P_{a-1}(x) - \frac{\sigma_2(a-1)}{a-1}\,x\right) + a - 5,$$

since $a^2 < \sigma_2(a) < 2\,a^2$. We observe that $P_a(x) - \frac{\sigma_2(a)}{a}\,x$ has non-negative coefficients. Finally, this implies that $F_a'(x) > 0$ for all $x > 1$. Thus, $F_a(x) > 0$ since $F_a(1) > 0$. □

Corollary 7.3. *Let $x \ge 1$ and $a \ge 3$. Then $\Delta_{a,1}'(x) > 0$.*

Remarks. (a) The real part of the zeros of $\Delta_{a,1}(x)/\,x$ is negative for $3 \le a \le 100$.

(b) The method of the proof is similar to the one outlined in [17].

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